Similarity Judgments Reveal the Multidimensional Representations of Naturalistic Faces

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Abstract

The human face affords a remarkable array of complex social cues, yet our understanding of how it is spontaneously perceived relies almost exclusively on controlled, posed stimuli that suppress the natural variation which makes faces so informative. Here, using a data-driven similarity task with 75 naturalistic face images, we mapped perceptual judgments across 365 adults aged 20 to 76. The results indicated that naturalistic face space was hierarchically organized by physical and demographic cues: smiling faces formed a coherent category that transcended gender and age, whereas non-smiling adult faces stratified along apparent gender. Multidimensional scaling and representational similarity analysis revealed that perceptual organization was driven by physical attributes – including facial action unit and age – rather than higher order social inferences or classic emotion categories. These findings reveal a robust, flexible geometry that accommodates naturalistic facial variation, advancing a more ecologically valid framework for real-world social perception.

Keywords: face perception; social cognition; emotions; social categorization; facial expressions

Research Transparency Statement

General Disclosures

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Preregistration:

The hypotheses, experimental methods, and analysis were preregistered (https://aspredicted.org/4P8_C88) prior to data collection. There were minor deviations from the preregistration (for details see Supplementary Table 1). All study materials, data, and analysis code are publicly available through Anonymous Github (https://anonymous.4open.science/r/multidimensional-DEF8).

Authorship:

N.P. played a lead role in project administration, data curation, formal analysis, investigation, visualization, writing–original draft, writing–review and editing, validation, an equal role in methodology, software and resources. A.R. played an equal role in software and resources, a supporting role in writing–review and editing and methodology. T.G. played a supporting role in methodology, software, and resources, and writing – review and editing. J.D.H. played a supporting role in conceptualisation and writing – review and editing. J.T. played a lead role in conceptualisation, funding acquisition, an equal role in methodology, resources, and a supporting role in writing – review and editing.

Multidimensional Representations of Naturalistic Faces

Making sense of the faces we see is an extraordinary feat. From a deceptively simple configuration of facial attributes, even the faces of strangers convey vast quantities of information, such as age, gender, identity, physiological state, personality traits, emotional appeals, and conversational intent (Buckland & Taubert, 2025; Cowen & Keltner, 2020; Holler, 2025; Leung et al., 2023; Liu et al., 2022; Scarantino et al., 2022; Sutherland & Young, 2022; White & Burton, 2022). To extract this multidimensional information, the brain is thought to compare incoming facial attributes against stored mental representations (Valentine et al., 2016). To isolate specific mechanisms, previous research has traditionally relied on tightly controlled face stimuli, varying attributes along isolated dimensions, such as identity, expression, gaze or face shape, while holding all other facial attributes constant. While such studies have provided foundational insights into how faces are categorized (Bruce et al., 1987; Bruce & Young, 1986; Porcheron et al., 2013; Rossion, 2002; Taubert et al., 2016), recognized (Taubert & Alais, 2011; Towler et al., 2023; White & Burton, 2022), elicit social inferences (Barrett et al., 2019; Fujimura et al., 2012; Russell, 1980), and guide predictions (Chanes et al., 2018), they necessarily restrict sampling to a curated subset of representational space. Consequently, these paradigms may not fully account for the variable and flexible feature integration characteristic of facial information 'in the wild'.

Further, traditional methods for eliciting behavioral judgments typically assess performance using categorization or recognition tasks, which make it difficult to elicit behavioral judgments without biasing responses toward predefined dimensions. For example, facial emotion recognition paradigms often rely on prototypical emotion categories (happiness, anger, fear, disgust, sadness, and surprise) represented by distinct configurations of facial action units (AUs) (Ekman & Friesen, 1978). Stimuli are typically controlled for these activations and paired with corresponding labels to assess behavior in laboratory settings. While highly valuable for establishing baseline mechanisms, this approach necessarily leaves aside non-prototypical AU configurations and constrains the diversity of descriptive

labels available to participants. Restricting response options in this way has been shown to influence performance, by potentially inflating consensus (Adams & Weisbuch, 2026; Nelson & Russell, 2013), limiting observations to predefined categories (Betz et al., 2019; Lindquist & Gendron, 2013), and/or privileging singular dimensions over latent, complex, or contradictory social cues (Matsumoto & Wilson, 2026). Because natural faces reflect multiple interacting dimensions, constrained approaches risk inadvertently reinforcing the theoretical commitments they intend to test (Peluso et al., 2025). Consequently, less is known about how mental representations of naturalistic faces are organized when facial signals retain information traditionally treated as noise.

More recently, data-driven methods have emerged for assessing perceptual face similarity in an unbiased way, making it possible to use larger and more complex stimulus sets which seek to explain face perception under more ecologically valid conditions. Here, our goal was to use one such method to characterize the multidimensional representations underlying a set of “uncontrolled” faces without imposing a facial attribute of interest. To this end, we quantified perceptual similarity using a triplet odd-one-out task previously used to understand latent dimensions in objects (Grootswagers et al., 2024; Hebart et al., 2020) with the Wild Faces Database (Peluso, Long et al., 2023), a diverse collection of ambient images displaying a range of heterogeneous facial gestures. The triplet odd-one-out task presents three stimuli simultaneously, with the participant choosing the one that is most different. No further instructions are provided to avoid biasing attention to any singular dimension, allowing the task to reveal similarities which might otherwise remain latent without presupposing or labeling the attributes of interest in advance.

In order to characterize the multidimensional space revealed by this paradigm, we separately collected behavioral ratings for 75 Wild Face images, consisting of 70 human faces (35 male, 31 female, four ambiguous) and five illusory faces (pareidolia), which we retained given evidence that perceiving face-ness in objects influences the perception of social cues such as valence and gender (Saurels et al.,

2024; Taubert et al., 2023; Wardle et al., 2022). These 75 faces were independently rated using traditional emotion category labels (happiness, sadness, anger, fear, disgust, surprise, neutral), and along twelve other dimensions previously identified as important for social cognition (Adolphs et al., 2016; Baudouin et al., 2025; Bruce et al., 1987; Lin et al., 2025; Peluso, Long et al., 2023; Norton & Pettegrew, 1979; Porcheron et al., 2013; Todorov et al., 2011). We included categories perceived from relatively stable facial attributes (chronological age, gender, gaze direction), and dimensions derived from more variable facial features such as affective inferences (valence, intensity, energy), and social traits (attentiveness, attractiveness, familiarity, genuineness, likeability, and trustworthiness).

We then used Representational Similarity Analysis (RSA), Multidimensional Scaling (MDS), and hierarchical clustering to map this representational space. These allowed us to directly compare participants' similarity judgments against our socio-evaluative ratings, low-level image properties using Gabor-Jets, VGG Face, CIE L*a*b* color and luminance (Grootswagers et al., 2019; Parkhi et al., 2015; Yue et al., 2012), and computationally derived descriptions of facial activity (OpenFace 2.0; Baltrusaitis et al., 2018)). Crucially, we demonstrate that naturalistic face similarity is driven by physical and structural face differences which do not conform to higher-order social inferences or classic emotion categories. The results provide an important step toward an ecologically informed model for describing how humans make sense of faces in the real world.

Materials and Methods

Participants

We recruited 365 healthy participants between 20 and 76 years of age (182 females, 1 non-binary; *M = 41.95, SD* = 13.64) via Prolific (www.prolific.com) to participate in an online experiment. To ensure we had enough power to examine whether representations would vary as a function of age, recruitment was streamed by nine age groups (20-24, 25-29, 30-34, 35-39 etc.) to ensure each group sampled a minimum of 40 participants containing an equitable distribution of gender. This number was

derived through pilot testing to ensure there were at least 20 observations for each pairwise similarity comparison (see Supplementary Text 1). Participants were excluded if they were previously diagnosed with neurological conditions. Given the risk of non-human agents participating in online experiments, we manually screened participants for implausibly fast task completion (< 20 minutes), and none were excluded. Participants were primarily based in the UK, Canada, and the United States and were paid approximately $21 AUS (£11 GBP) for their participation, for which they provided informed consent. The study was approved by The University of Queensland's Human Research Ethics Committee (2024/HE000198).

Stimuli

Stimuli comprised 75 images (hereafter referred to as the WFD-75) selected from the Wild Faces Database using a computational Variational Interpretable Concept Embeddings (VICE) model (Muttenthaler et al., 2022). This model previously identified 15 dimensions underlying triplet odd-one-out judgments across 900 WFD images, excluding animal faces (Taubert et al., 2024). To ensure broad coverage of facial attribute space, the top five exemplars from each dimension were selected comprising 70 human faces (35 male, 31 female, four ambiguous) and five objects with face-like qualities (pareidolia). Images were 400 x 400 pixels and varied widely in head orientation, gaze direction, color profile, and image quality.

Procedure

Triplet Odd-One-Out Task

Visual similarity judgments were acquired remotely using a triplet odd-one-out task previously used to examine representational geometry in natural objects and human actions (Bockes et al., 2025; Grootswagers et al., 2024; Hebart et al., 2020). The task was programmed in jsPsych (de Leeuw, 2015) and hosted on Pavlovia (pavlovia.org). Each trial displayed a random combination of three faces arranged equidistant from each other from the center of the screen (Fig. 1). Participants were asked to

select one image as the odd-one-out, as being the image they judged to be most different to the other two. No additional instructions were provided to avoid biasing to particular features, and each triplet remained visible until a choice was made by responding with a mouse click on the chosen image, initiating a 500 ms interval before the next trial. Breaks were provided every 201 trials, with participants completing 500 trials yielding a total of 182,500 similarity judgments across the entire experiment.

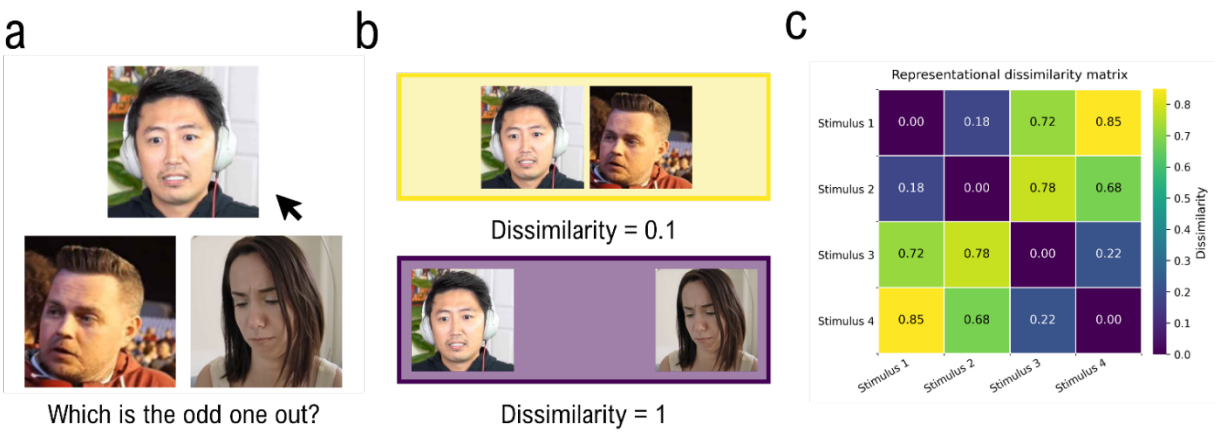


Fig. 1.

Trial visualization of the triplet odd-one-out task. Sample images are from 'A database of heterogeneous faces for studying naturalistic expressions', by H. Long, N. Peluso, C.I. Baker, S. Japee, & J. Taubert, 2023, *Scientific Reports, 13*(1), 5383, https://doi.org/10.1038/s41598-023-32659-5. Openly available under the CC-BY licensing agreement. Participants were asked to judge the odd-one-out (a), and we used pairwise judgments to calculate dissimilarity between each pair of images (b), where a low dissimilarity score (top image pair) indicates they were perceived as similar when presented together in a trial, and a high dissimilarity score (bottom image pair) indicates they were perceived as more different. Dissimilarity matrices (c) were constructed using these scores to examine covariation between similarity judgments and candidate models.

Socio-Evaluative Ratings Task

To characterize visual information in the WFD-75, independent ratings were collected from a separate sample of 75 undergraduates (age 17-25; 47 females, 27 males, 1 undisclosed) recruited via The University of Queensland research participation scheme, who volunteered in exchange for course credit. Participants were asked to rate each image along dimensions of attentiveness, attractiveness,

energy, familiarity, gaze, and trustworthiness using a continuous Likert sliding scale from 1 on the left to 9 on the right (for example, “How likeable/attractive/trustworthy is this face?”; 1 = Not at all, 9 = Extremely), before advancing to the next image. Gaze ratings were obtained with the question “Which direction is the face gazing? (Report the direction from your perspective. If the eyes are closed, please guess)” with 1 = To the left, 5 = direct, 9 = To the right. Chronological age ratings were obtained using a continuous slider from 0 – 100. Between 25 and 42 judgments were collected for each rating and the study was approved by The University of Queensland’s Human Research Ethics Committee (2024/HE000198). Additional ratings for gender, valence, emotion category, intensity, and genuineness were obtained from previous studies described elsewhere (Peluso, Long et al., 2023; Wei et al., 2026, see Supplementary Text 2).

Analysis

Representational Similarity Analysis (RSA)

Behavioral data were analyzed in MATLAB 2025b using the Representational Similarity Analysis (RSA) framework, a multivariate analysis used in previous tasks to characterize patterns of responses to visual stimuli (Kriegeskorte et al., 2008). Odd-one-out judgments were used to calculate dissimilarity between each pair of experimental stimuli (Fig. 1). For each pair of faces, similarity was computed as the proportion of trials where the pair appeared together and neither image was selected as the odd one out, divided by the total number of trials that included that pair. These similarity scores were then subtracted from 1 for conversion into dissimilarity scores, such that higher values indicated pairs were perceived as more different. Dissimilarity scores were calculated for all pairs of 75 stimuli, resulting in a 75 x 75 Representational Dissimilarity Matrix (RDM) comprising 2,775 unique pairs.

We also used RSA to compute the split-half consensus of similarity judgments across participants by randomly dividing participants into two groups and correlating their RDMs across 1,000 bootstrap iterations. Using the same bootstrap method, we also assessed between-trial reliability by

dividing trials into odd- and even-indexed halves, retaining the full *N* in each half. Correlations were corrected using the Spearman-Brown formula.

Model Representational Dissimilarity Matrices. We built multiple model RDMs for perceived age, attentiveness, attractiveness, energy, familiarity, gaze, gender, genuineness, intensity, likeability, trustworthiness and valence by computing pairwise dissimilarity as the absolute value of the difference between the two average rating scores using Euclidean distance. The emotion category RDM was constructed using dissimilarity in the rated emotional profile of each face. First, for all 75 faces we constructed a 9 x 1 vector based on the frequency with which each emotion category label (anger, fear, disgust, sadness, neutral, happiness, surprise, other or unknown) was selected for each face in a previous labeling task (Peluso, Long et al., 2023). These vectors were correlated across all pairs of stimuli using Spearman's Rho and the resulting Rho value subtracted from 1 to derive the emotion category RDM.

Computational image-derived model RDMs were built using numerical vectors that described image luminance and color distribution (both using CIE L*a*b* color space), other image features extracted from Gabor-Jets (Yue et al., 2012) and VGG Face (Parkhi et al., 2015), and facial action unit intensity detection using OpenFace 2.0 (Baltrusaitis et al., 2018). These models allowed assessment of the low-level and facial movement features previously found to correlate with visual similarity. For exact model specifications see Supplementary Text 2.

Statistical Inference Using Multiple Regression on Distance Matrices

We used multiple regression on RDM distance matrices to estimate each candidate model's unique contribution to behavioral similarity with all other models held constant. The behavioral RDM for all participants was vectorized and regressed simultaneously onto the corresponding vectors from each of our candidate models as predictors, yielding standardized regression coefficients (β) reflecting each model's unique contribution after partialling out shared variance. One challenge with inference using

RSA is that observations within an RDM are not independent. We therefore followed a stimulus-label randomisation approach (Kriegeskorte et al., 2008; Nili et al., 2014) and combined this exchangeability method with the Freedman-Lane procedure (Winkler et al., 2014), under which each predictor is tested against a null that retains the effects of the remaining predictors. Permutation *p*-values were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate. Analysis was conducted using RStudio 2026.01.1+403.

Multidimensional Scaling (MDS)

We used Multidimensional Scaling (MDS) to map the underlying structure of the behavioral RDM, such that faces perceived as more similar are positioned closer to each other in space than faces perceived as more different (Kruskal & Wish, 1978). To assess the optimal number of dimensions, we computed MDS solutions from 2 – 8 dimensions using Kruskal's Stress 1 formula to measure fit quality, achieving stress 0.077 by seven dimensions but with no clear elbow (see Supplementary Fig. 1). Given improvement dropped to ~1.2% between seven and eight dimensions, we retained the seven-dimensional (7D) solution as an optimal balance between preserving meaningful structure while retaining interpretability.

Hierarchical Clustering Analysis

We applied hierarchical clustering (average linkage) using the pairwise Euclidean distances derived from the 7D MDS embedding using MATLAB 2025b. Average linkage defines the similarity between two clusters as the average distance between all possible pairs of observations across the two groups. Given naturalistic face stimuli are more perceptually graded, this algorithm was chosen for its robustness to outliers and to preserve the geometric structure of the underlying pairwise distances. We evaluated the optimal number of clusters by computing mean silhouette width for solutions from $k = 2$ to 10, using Euclidean distances in the same seven-dimensional space in which the clustering was performed. Silhouette width was maximal for the trivial two-cluster solution ($\bar{s} = .35$), which separated

the five pareidolia stimuli from the remaining faces, while in the range of human faces silhouette width rose from a minimum at k = 3 ($\bar{s}$ = .20) to a maximum at k = 7 ($\bar{s}$ = .29), with k = 8 closely comparable ($\bar{s}$ = .28) before plateauing to k = 10 ($\bar{s}$ = .28). Given the graded nature of the silhouette values, we also examined the merge heights of the dendrogram itself, where two face images high in similarity are located in close proximity and connected by vertical branches which are low in height. Excluding the human/pareidolia cluster split (gap = .185), the largest increases occurred at k = 4 (gap = .055) and k = 8 (gap = .044); no other solution exceeded .025. We therefore converged upon an eight-cluster solution which explained 66.59% of the total variance in the 7D MDS space, representing a robust clustering solution with strong overall separation. Clusters were profiled using the independent ratings data and OpenFace 2.0 using RStudio 2026.01.1+403.

Results

Naturalistic faces elicit robust consensus in similarity judgments across the adult lifespan

We quantified consensus among participants using split-half reliability and found similarity judgments were remarkably consistent both across participants (r = .968) and across trials (r = .969), Spearman-Brown corrected. Given the equitable distribution of female and male participants within the full sample (n = 182 each), we also quantified consensus within and between each gender group which we found to be similarly high (females: across-participants, r = .938 and across-trials, r = .938; males: across-participants, r = .936 and across-trials, r = .939). Similarly, the behavioral RDMs that were generated by male and female participants were highly correlated with each other (Spearman's rho r = .92, $p < .001$).

To investigate any potential age differences in similarity judgments, we also verified consensus by grouping data by participant age (20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64+) to construct nine separate behavioral RDMs. Each age RDM contained 40-43 participants, each covering 100% of 2,775 pairwise comparisons with a median count of 21-23 observations. Despite the relative

sparsity in these RDMs compared to the full sample RDM (which contained a median count of 149-197 observations per pair), consensus remained strong with all age groups significantly correlated with each other, ranging between r_s = .697 and .764, p < .001, FDR corrected.

Physical features, not social inferences, predict naturalistic face similarity

After establishing impressive agreement between participants as to which faces were more similar than others, we next sought to identify which of our socio-evaluative ratings and stimulus-derived computational models might independently predict these judgments using multiple regression. Given our task included illusory faces (i.e., pareidolia) which do not feature facial action units and may contain contrasting visual statistics to human faces (such as luminance and color), we conducted three separate analyses: the first included all 75 stimuli (70 human faces and 5 pareidolia) with 17 predictors, excluding OpenFace. The second introduced OpenFace and excluded six stimuli OpenFace could not process (five pareidolia stimuli and one black-and-white human face) yielding a 69-face dataset with the full 18-predictor set. Given Analyses One and Two differed in both stimulus set and predictor set, Analysis Three was fitted on the 69-face dataset with the same predictors as Analysis One to be able to compare the effect of stimulus removal separately to the inclusion of OpenFace. Overall model fit in all three analyses was significant (permutation p < .001).

In Analysis One (75 stimuli, all predictors except OpenFace; R^2 = .117), we found age (β = .196), color (β = .158), and familiarity (β = .143) each contributed independently to behavioral similarity after FDR correction (Fig. 2, orange lines). In Analysis Two (69 human faces, including OpenFace; R^2 = .102), only age (β = .217) and OpenFace (β = .202) were significant, suggesting the contribution of color and familiarity in Analysis One may have been driven by the perceptual distinction between human and illusory face stimuli (Fig. 2, green lines). To better assess the effect of stimulus set before the introduction of OpenFace, Analysis Three (69 human faces, all predictors except OpenFace; R^2 = .063) revealed that only age (β = .208) and luminance (β = .104) contributed unique variance (Fig. 2, blue

lines). That luminance survived pareidolia removal but was absorbed by OpenFace suggests facial AU intensity estimates may carry some luminance dependence. Together, these results converge on physical differences between faces, such as apparent age and facial action units, as meaningful predictors of naturalistic face similarity. Importantly, once these were controlled, we found no contribution from any of our other socio-evaluative predictors (see Supplementary Tables 2 – 4).

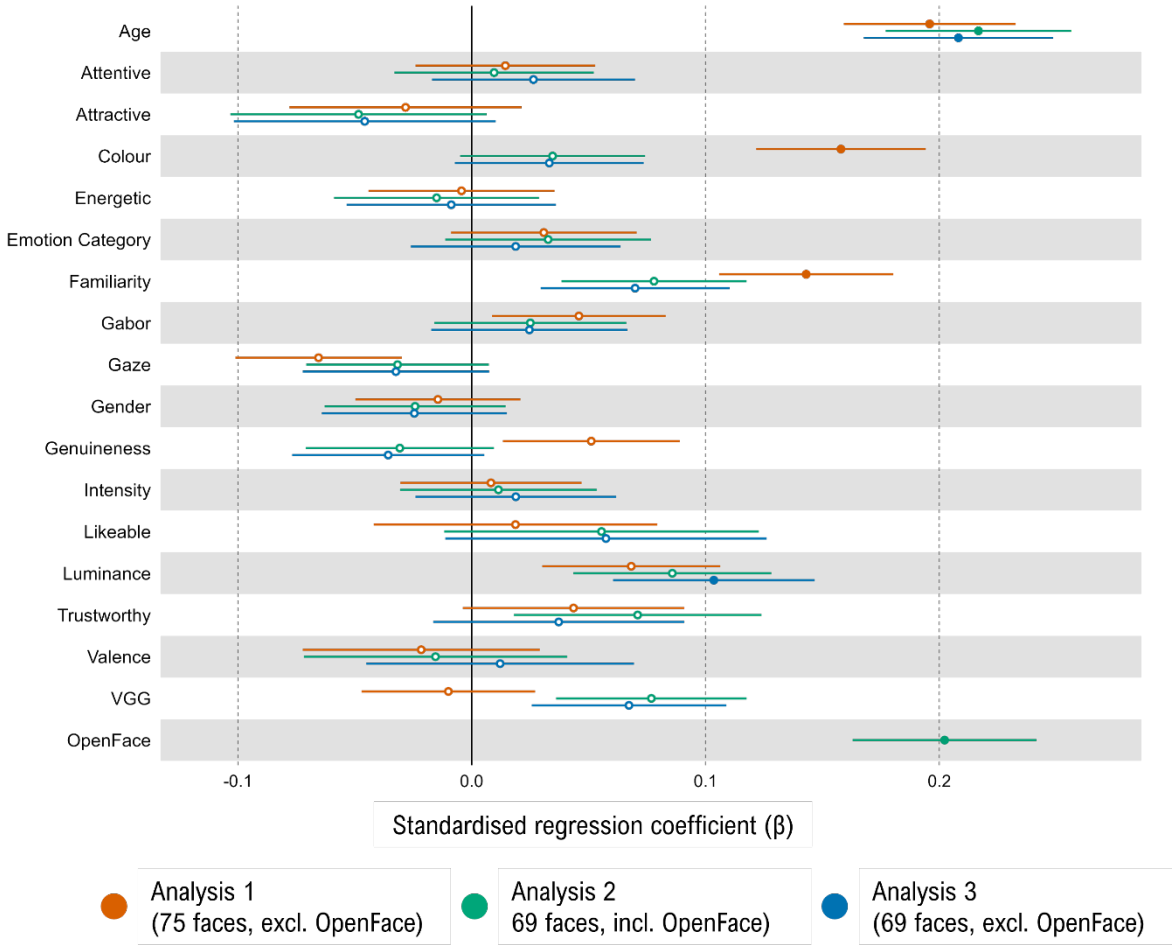


Fig. 2.

Predicting representational similarity judgments using independent candidate models. Points indicate standardized regression coefficients; bars indicate ±1.96 ordinary least squares standard errors for descriptive purposes and may exclude zero for a predictor that is not significant under permutation. Filled markers indicate predictors surviving stimulus label permutation and Benjamini–Hochberg false discovery rate correction ($q < .05$); open markers indicate non-significant predictors.

Hierarchical clustering reveals interpretable organization of naturalistic face space

Our results so far reveal naturalistic faces elicit high-dimensional and remarkably consistent similarity judgments, which are explained only in part by attributed age and facial AU intensity. To identify the face attributes underlying these dimensions, we used hierarchical clustering as a means of recovering categorical structure while preserving dimensionality, grouping stimuli solely on the basis of their pairwise dissimilarities. Figure 3 displays the full clustering solution as a vertical dendrogram alongside the RDM arranged in the order of images derived from hierarchical clustering, revealing highly

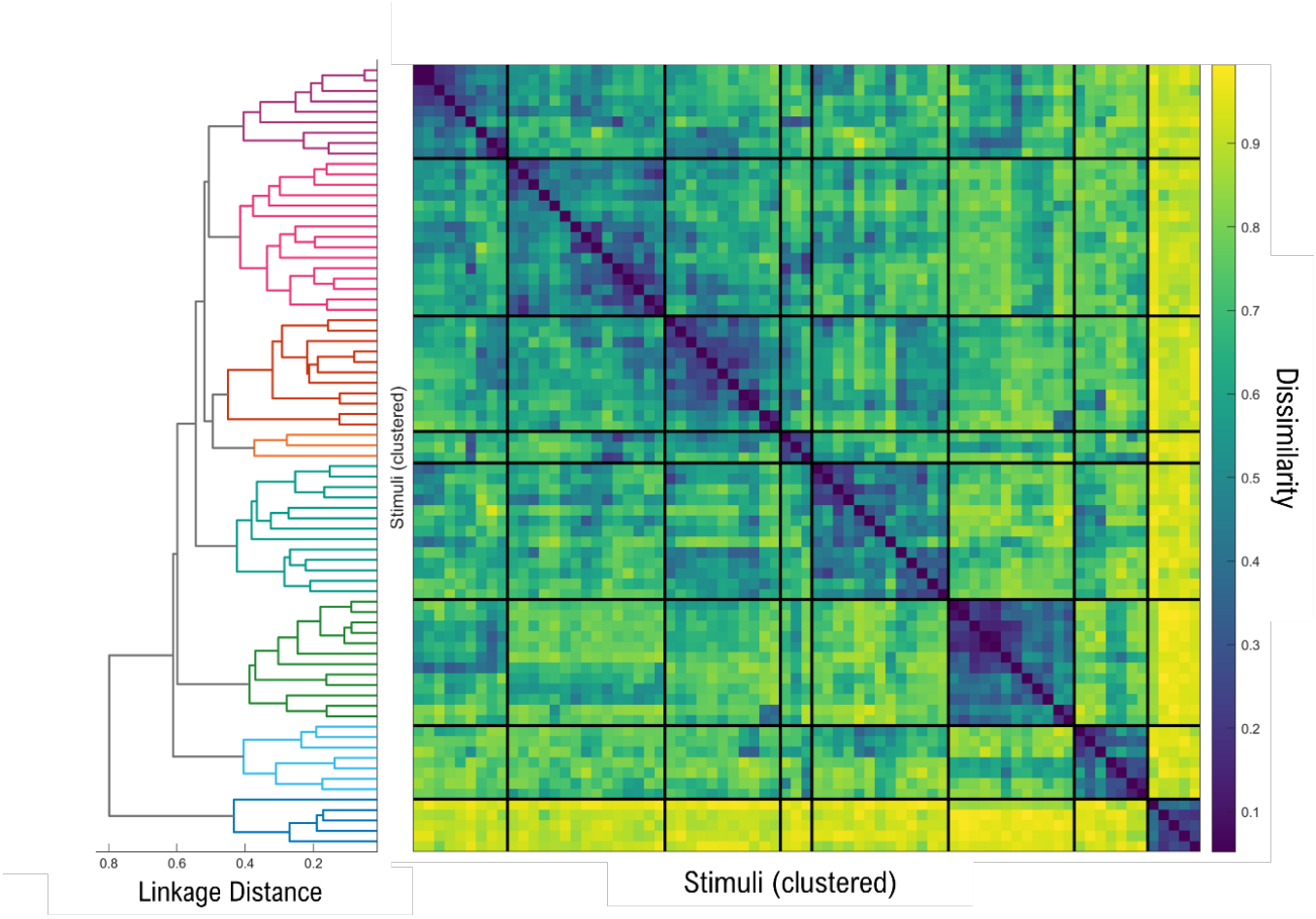


Fig. 3.

Representational dissimilarity matrix (RDM) for all participants arranged in dendrogram order after hierarchical clustering. The vertical dendrogram is displayed on the left, with the highest distance cluster at the bottom. Black lines on the RDM show the boundaries between clusters (k = 8, see Methods).

distinct clusters of low dissimilarity along the diagonal (dark blue). Pareidolia images (*n* = 5) emerged as the most distinct cluster and were judged as strongly dissimilar to human faces, visible as a high dissimilarity (yellow) horizontal stripe along the bottom of the RDM.

Accordingly, if the resulting clusters are interpretable, then face images should share certain identifiable properties that might explain category membership. In Figure 4, we inspected the face images within each cluster alongside the dendrogram solution, and found each cluster contained interpretable, yet complex combinations of visually similar stable and dynamic social cues. We describe these groups in terms of their shared features.



Fig. 4.

Hierarchical clustering of similarity judgments for all participants revealed eight clusters. Displayed vertically, the dendrogram (left) is ordered by linkage distance from most distinctive clusters (top) to least distinct (bottom). Images correspond to the face images comprising each cluster and displayed in the same order as the resulting dendrogram, such that faces proximal to each other were perceived as more similar.

The most distinct cluster (*n* = 5, blue) contained all face pareidolia images (Fig. 4, top), indicating a strong categorical boundary between human faces and illusory faces in objects which emerged at the highest branching point of the dendrogram (~0.8). Among human faces, the most distinct cluster contained images of children/babies (*n* = 7, cyan) irrespective of variability on other dimensions (e.g. gender, facial gesture). Indeed, mean attributed age for this group of stimuli was 5.67 (*SD* = 1.84) in comparison to the mean age of the remaining stimuli (which ranged from 21.86 - 39.14). Interestingly, older faces did not similarly form a unique cluster, but were distributed into other adult clusters, suggesting an asymmetry in how age information is combined with other social cues. Cluster three revealed a uniquely coherent category for smiles (*n* = 12, green) which transcended variations in stable demographic cues such as gender, ethnicity, and age, and highly variable cues such as gaze direction and head orientation. Remaining clusters were organized by salient non-smiling facial gestures (*n* = 13, teal), faces with hands covering mouths (*n* = 3, orange), or by gender and/or ethnicity (*n* = 11, red; *n* = 15, vermillion; *n* = 9, purple). Interestingly, while the perceived gender and ethnicity of face stimuli seemed robust organizers of clusters six to eight, these cues also seemed important within clusters two to five, with faces from the same gender or ethnicity tending to be proximal to each other.

Demographic and Affective Dimensions Form Distinct Spatial Gradients

To understand how dimensions were distributed across the global embedding as well as within each cluster, we first mapped the coordinates derived from the seven-dimensional solution onto two-dimensional multidimensional scaling (MDS) solution and colored each stimulus in the embedding according to its cluster membership (Fig. 5). We then repeated this process with the face stimuli themselves. This revealed that each cluster was spatially coherent and well differentiated even in this low dimensional simplification.

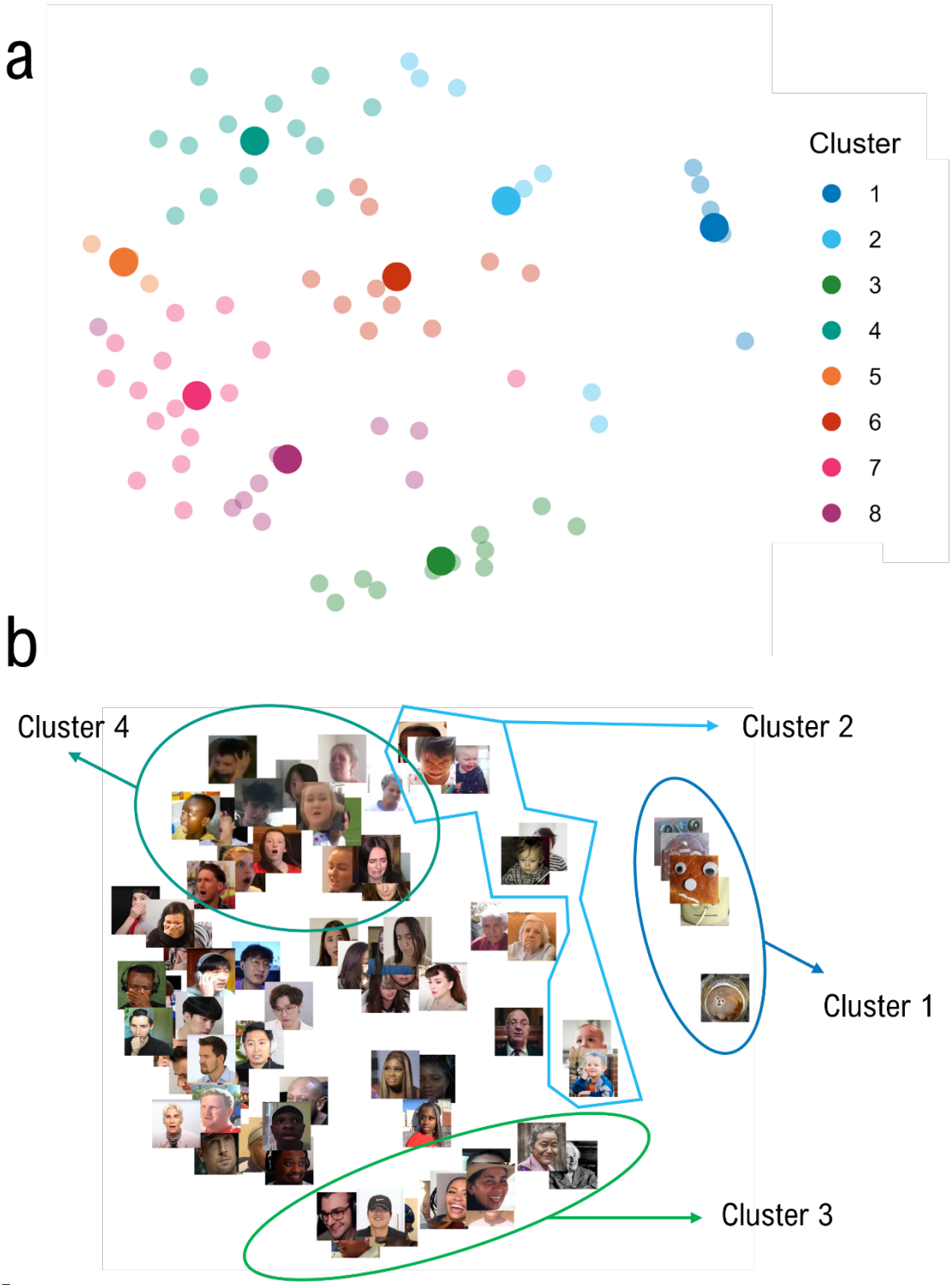


Fig. 5.

Two-dimensional mapping from high dimensional similarity judgments. The two-dimensional multidimensional scaling (MDS) solution derived from similarity judgments and mapped according to (a) the eight clusters derived from the seven-dimensional solution projected onto the same two-dimensional coordinates, and (b) to face stimuli with the first four clusters defined.

We then used the same method to examine how individual dimensions were spatially organized across the same embedding by recoloring each stimulus according to our ratings of perceived age, gender, and valence (Fig. 6). Here, each face image is represented as a colored dot on the embedding according to its unstandardized value on each dimension and presented alongside the same values visualized as boxplots according to cluster. By doing so, we found that cluster specificity in each dimension mapped to distinct spatial gradients across the global embedding. First, while age evaluations were noticeably younger in cluster two (*M*age = 5.67, where age was rated from 0 – 100) these images did not cluster together when mapped in two-dimensional MDS space, but rather were distributed along a vertical gradient (Fig. 6a). Intriguingly, older faces were not located directly opposite the youngest faces on the embedding, but rather adjacent in a similar vertical distribution. Similarly, mean gender ratings per cluster (Fig.6b) confirmed the highest gender specificity in clusters six (*M*gender = 2.53, where 1 = extremely feminine) and cluster seven (*M*gender = 7.56, where 9 = extremely masculine), yet the embedding revealed a distinctly different spatial organization with feminine faces vertically concentrated in the center, masculine faces to the left and androgynous faces to the right (Fig. 6b). Valence ratings (Fig. 6c) also revealed strong cluster-specific alignment, particularly in cluster three (smiles, *M*valence = 7.2) where highly positive faces formed a dense, well-defined region in MDS space. At the opposite extreme to cluster three, cluster four contained face images rated as more negative (*M*valence = 3.27) confirming the presence of a graded valence axis through MDS space, with the center however remaining more diffuse.

Using the same method, we further discovered that arousal-related dimensions of energy and intensity were not independent of valence (Fig. 7). Both scatterplots and recolored MDS plots showed a similar V-shape pattern across the embedding for both energy and intensity dimensions, with lowest values in face pareidolia and rising toward both the high-valence region (smiling faces, cluster three) and

the low-valence region (cluster four). See Supplementary Table 5 for all socio-evaluative ratings by cluster, Supplementary Fig. 2-4 for boxplots and recolored MDS plots for all socio-evaluative ratings.

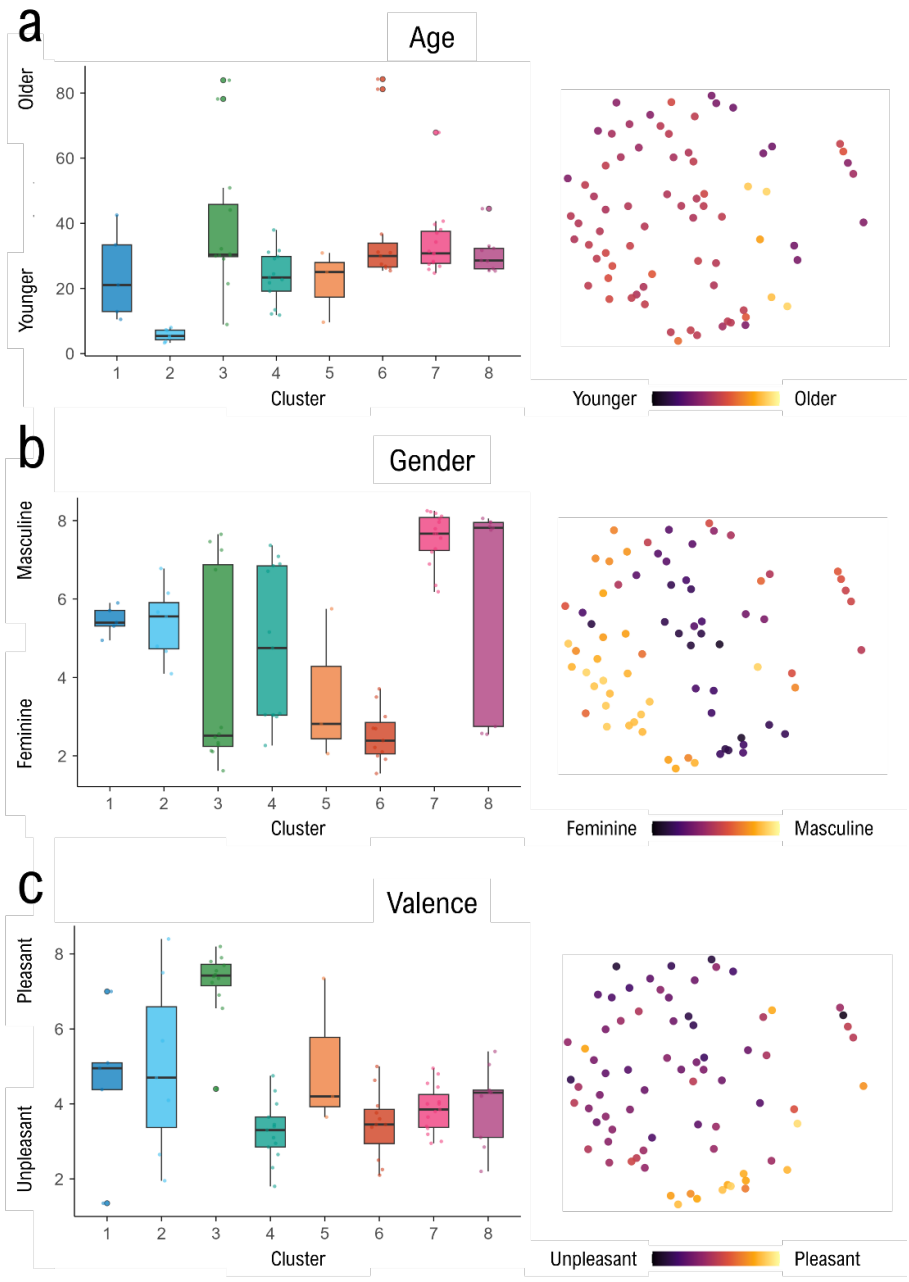


Fig. 6.

Cluster profiles by age, gender, and valence. Boxplot panels display the mean unstandardized ratings for individual face stimuli (jittered points) and their interquartile range (box) according to cluster for (a) age, (b) gender and (c) valence. Alongside, the same ratings were standardized for individual face stimuli before projection onto two-dimensional coordinates derived using the seven-dimensional MDS solution.

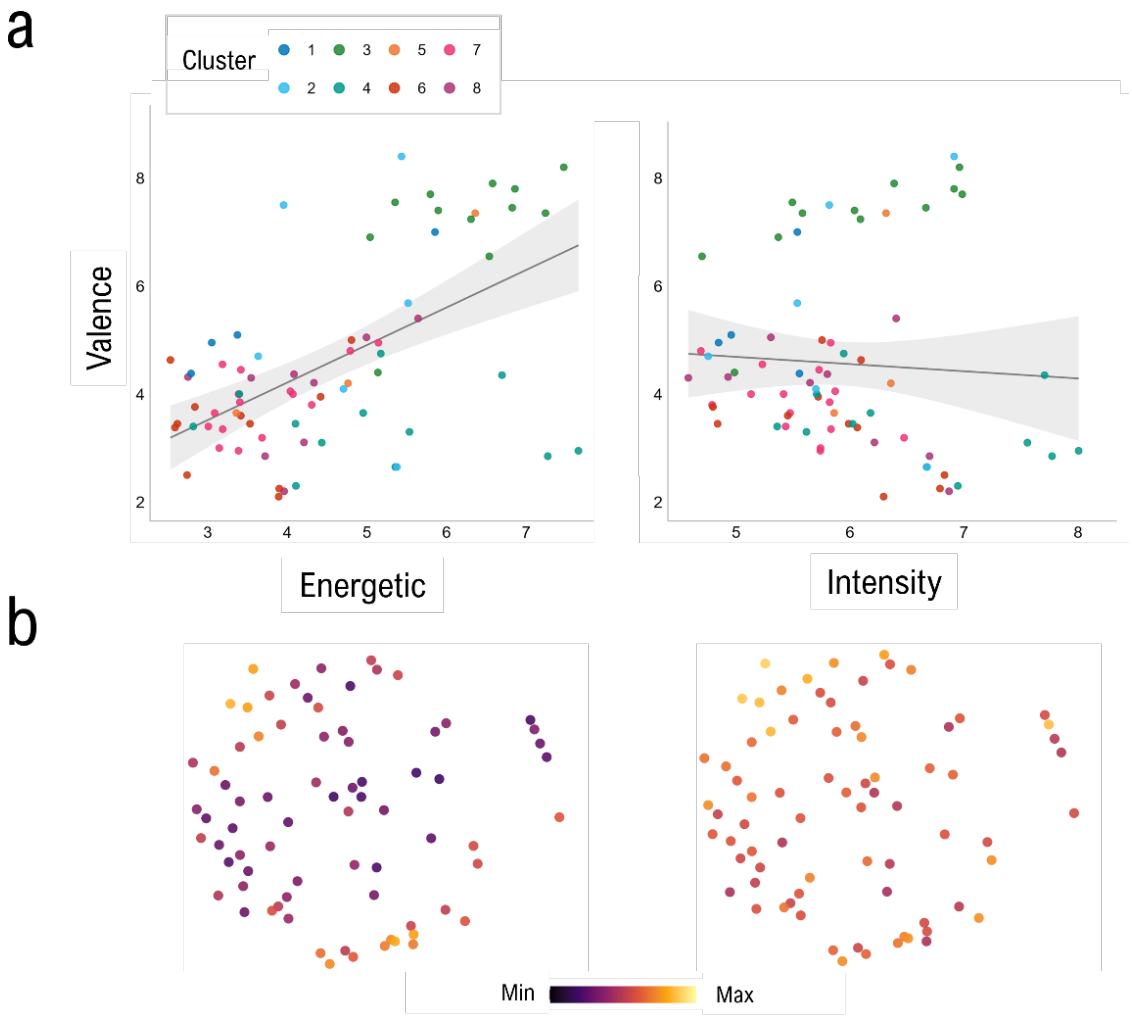


Fig. 7.

Relationship between valence, energy and intensity in naturalistic face space. Scatterplots (a) with linear regression lines and 95% confidence intervals (shaded regions) mapping unstandardized mean rating correlations between (left) valence and energy and (right) valence and intensity ratings; individual data points are colored by cluster. The bottom panels (b) show the seven-dimensional MDS solution projected onto two-dimensional coordinates with face stimuli recolored according to their standardized mean ratings of energy (left) and intensity (right).

High Heterogeneity in Emotion Category Labels and Action Units Within Clusters

Finally, given our finding that OpenFace was a significant predictor of continuous similarity, it was also possible that our clustering was indexing a measure of prototypical emotion categorization, either via emotion label attributions, or facial AU configurations thought necessary for emotion category recognition. To investigate, we first visualized the percentage of face images for each cluster that corresponded to each of nine emotion category labels (anger, disgust, fear, happiness, sadness, surprise, neutral, other, or unknown; Fig. 8a). This revealed that clusters were not associated with coherent

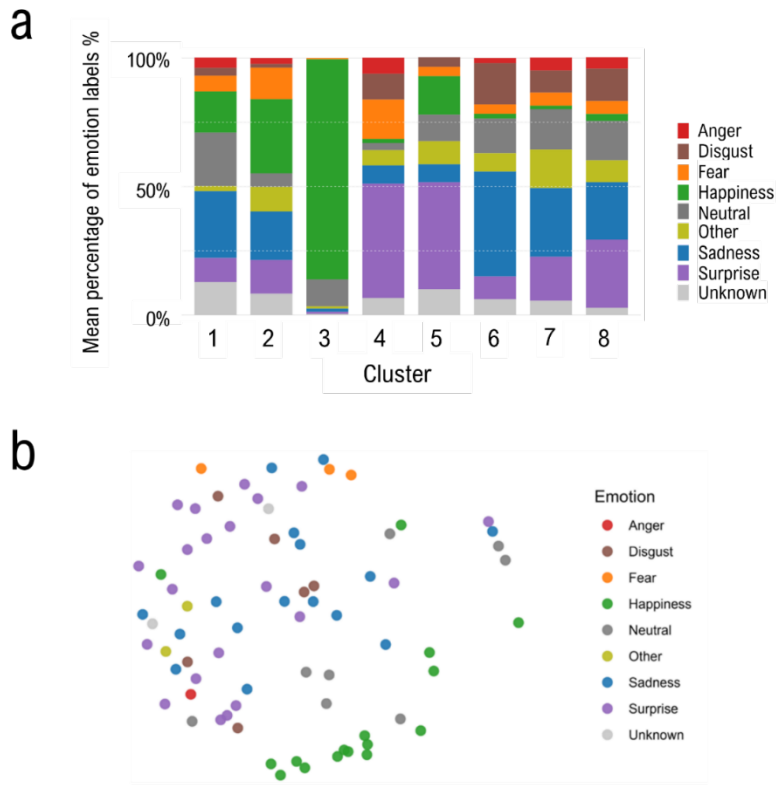


Fig. 8.

Emotion category distribution in naturalistic face space. Each cluster is visualized (a) according to the percentage of raters who assigned a given emotion label to stimuli within each cluster, and (b) across the seven-dimensional MDS solution projected onto two-dimensional coordinates, recolored according to the emotion category most frequently assigned to each face.

emotion category labels, with the remarkable exception of cluster three (smiles) which showed near-exclusive attributions for happiness. This specificity was also clear when emotion category labels were mapped to MDS coordinates (Fig. 8b).

We then asked whether this emotion category heterogeneity extended to facial action unit (AU) detection profiles by calculating the coefficient of variation (CV = *SD/M*) from the 69-stimuli OpenFace AU intensity detections across cluster members (Fig. 9) as well as the proportion of cluster members with active AUs and the mean intensity for each AU by cluster (Supplementary Fig. 5 - 6). Again, we found evidence of considerable variability within clusters, characterized by predominately high CV values (1.0–2.8). While cluster three (smiles, *n* = 12) was notably coherent across both AU12 (Lip corner puller, CV = 0.5) and AU06 (Cheek raiser, CV = 0.5) – activations thought crucial for so-called genuine or Duchenne smiles - cluster five (*n* = 3) also revealed similar specificity across several mouth-related AUs including AU14 (Dimpler, CV = 0.5), AU07 (Lid tightener, CV = 0.6), and AU20 (Lip stretcher, CV = 0.3) despite these images all containing faces with hands covering the mouth region. This confusability suggests challenges remain in detecting AUs in naturalistic face stimuli, and that naturalistic face stimuli may contain more complex combinations of AUs than basic emotion category models predict.

Discussion

In this study we combined an unbiased measure of perceived similarity, data-driven multivariate analyses, and independent perceptual and socio-evaluative explanatory models, to map the similarity across 75 naturalistic face images, allowing the stimuli to freely vary without specifying in advance which dimensions should matter. Three major findings emerged. First, we found impressive agreement in how observers spontaneously responded to highly variable naturalistic facial attributes across the adult lifespan, even without providing explicit instructions. Second, we found multiple categorization strategies which flexibly adapted to distinctive physical differences between faces; facial cues consistent

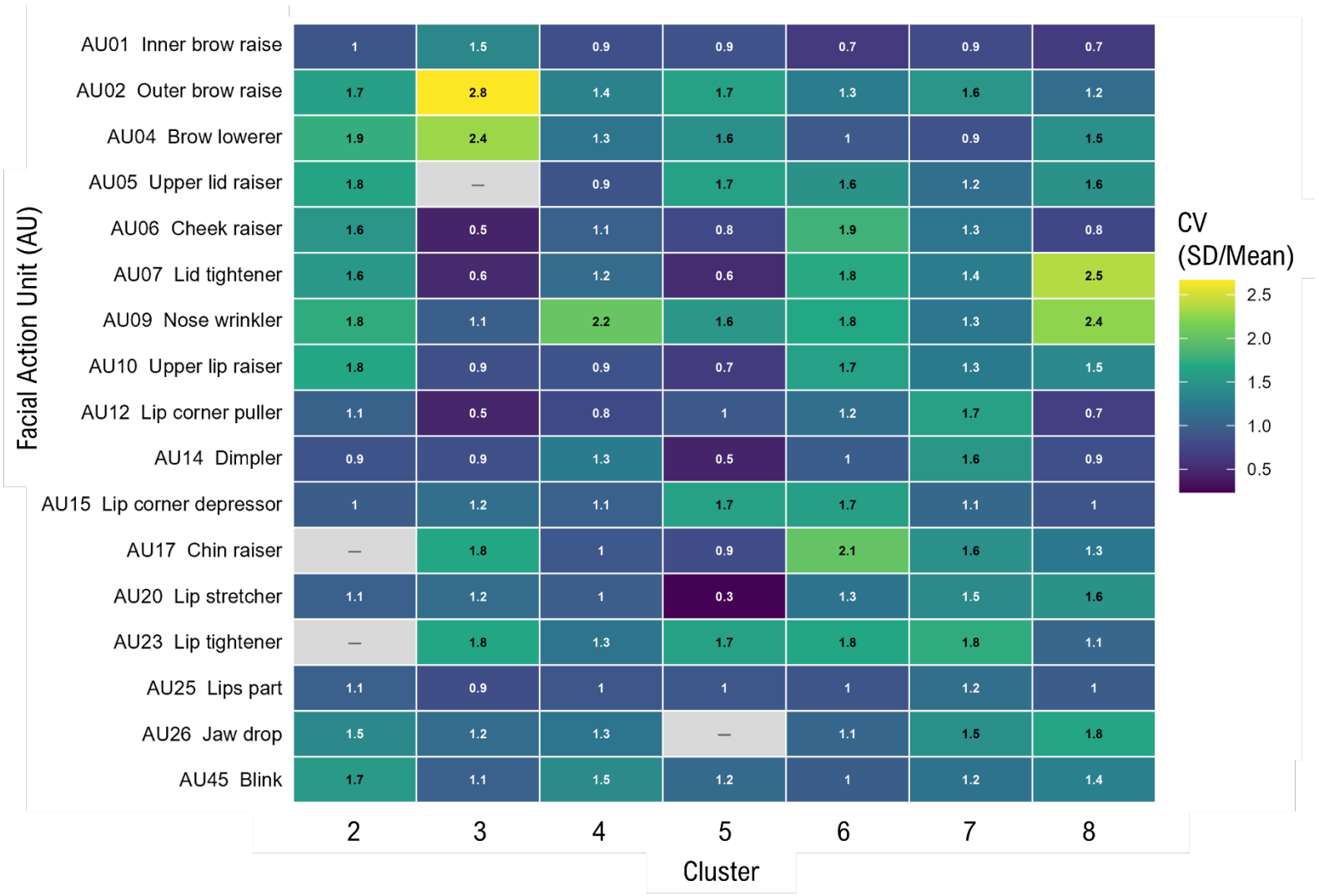


Fig. 9.

Heterogeneity in facial action unit (AU) intensity profiles across clusters (excluding cluster 1, pareidolia). Each cell displays the coefficient of variation (CV = SD/mean) for AU intensity within each cluster, computed from OpenFace 2.0 detections across cluster members. Higher CV values (yellow) indicate greater within-cluster variability in AU intensity; lower CV values (dark purple) indicate greater consistency. Gray cells (—) represent undefined AUs due to near-zero mean intensity (mean intensity < 0.05) for that cluster.

chronological age, gender, and ethnicity provided pronounced hierarchical structure, yet smiles were elevated to cut across these stable demographic cues. Third, we found multidimensionality varied both within clusters and along core dimensions such as valence, yet these patterns were not easily reducible to basic emotion categories. Together, these findings reveal a physically grounded, yet flexible geometry

for incorporating multiple behaviorally relevant social cues in naturalistic faces, with important consequences for understanding the foundations underlying human face perception.

Crucially, we demonstrate that this representational space was not explained by many of the socio-evaluative dimensions that have dominated person perception research, such as classic emotion categories, trustworthiness, attractiveness, likeability, genuineness, or even valence. This implies that when stimuli are allowed to vary and tasks remain unconstrained, observers anchor their judgments primarily to physical differences in facial attributes. However, these same physical differences resist compression into inferential categories which may appear more dominant in highly controlled stimuli. This asymmetry is notable given that many of the evaluative dimensions we tested are themselves thought to arise from configurations of the very features that predict similarity (Adolphs et al., 2016; Todorov et al., 2011). Demonstrating that evaluative judgments can be recovered from facial features does not necessarily entail that those judgments organize perception, and our results suggest the two questions are dissociable under naturalistic conditions. Therefore, a productive next step may be to ask how feature space similarity affords multiple, context-dependent inferences, rather than treating complex inferences as reducible to the feature configurations from which they can be predicted.

Yet any such account must contend with a feature space in which facial AU activity appears less orderly than traditionally described. While facial AUs were significant predictors of naturalistic face similarity, the activations themselves were considerably heterogeneous—even between faces that were judged as most similar. This variability was not entirely indiscriminate, however; AU06 and AU12 activations were markedly more consistent in smiling faces, indicating that OpenFace could detect coherence when coherence was present. That smiles consistently emerged as an anchor therefore suggests their visual salience is not merely an artifact of category-sparse task designs, but reflects a genuinely distinctive, unambiguous cue that observers spontaneously prioritize even amidst complex naturalistic variation (Peluso et al., 2025). However, we also found confident detections of mouth-

specific AUs in response to faces with no visible mouth at all (e.g., hand-over-face stimuli), highlighting current challenges in computational AU detection in the wild. More broadly, while facial action units remain an important objective measure of facial movement, our results indicate that more work is needed to understand how, when, and in what combinations they activate under unconstrained visual conditions.

More speculatively, our finding that naturalistic facial attributes are flexibly and hierarchically combined also raises the possibility that cues supporting the recognition of face familiarity may be less separable from those supporting the perception of facial gestures, not only when distinguishing faces we know from the faces of strangers, but in distinguishing the same person across multiple varying instances (Burton, 2013; Towler et al., 2023; White & Burton, 2022). Previous work has identified a set of shared facial attributes essential for processing familiar and unfamiliar faces, and has shown that OpenFace effectively captures these attributes (Abudarham et al., 2019; Tsantani et al., 2021). That AU intensity emerged as one of only two predictors to explain unique variance in similarity judgments is convergent with the view that these measurements index features relevant to both. Establishing whether they do so in naturalistic faces by manipulating familiarity directly is an important next step for understanding how observers learn to generalize and differentiate between faces outside the laboratory (Zhou et al., 2022).

Our approach is not without limitations; we make no claim that our stimulus set represents the full spectrum of naturalistic facial attributes, and a different set of ambient faces could produce a distinct representational topography. However, we intentionally chose a compact, highly diverse stimulus set in contrast to data-driven approaches operating at much larger, sparser scales (Bockes et al., 2025; Hebart et al., 2020). This design facilitates future fMRI and E/MEG investigation, enabling dense repeated sampling of the same images across different populations to isolate contextual effects and individual differences that remain understudied (Hehman et al., 2017; White & Burton, 2022).

Accordingly, future work should prioritize testing non-Western cohorts and scaling this paradigm to larger stimulus sets to determine how broadly this representational geometry generalizes.

Beyond advancing psychological theory, mapping how humans process unconstrained facial attributes matters directly for real-world applications. In computational settings, automated algorithms trained on prototypical face datasets routinely struggle under more ambient conditions; grounding artificial intelligence and computer vision models in naturalistic multidimensional representational geometries can guide more robust, generalizable facial analysis tools. This has important downstream consequences for mitigating bias in human and automated decision-making across security, legal, or recruitment settings. Further, in clinical settings, conventional measures of social cognition present images with static, prototypical facial gestures, typically with semantic or forced-choice tasks, penalizing observers for failing to map complex or ambiguous gestures onto predefined categories. Establishing benchmarks grounded in naturalistic, unconstrained facial space offers a more ecologically valid approach for capturing social cognitive difficulties that might be expected to occur in actual, daily life (Henry et al., 2023).

Taken together, we demonstrate that humans spontaneously use both broad categories and fine-grained dimensions to evaluate naturalistic faces. By advancing a new methodological framework that accommodates variation in both facial information and observer behavior, this work moves beyond traditional stimulus rewilding (Dawel et al., 2025; Long et al., 2023; Peluso et al., 2025) toward data-driven paradigms that allow the complex, flexible structure of face representations to guide more veridical models for describing how faces invite real-world meaning.

References

Abudarham, N., Shkiller, L., & Yovel, G. (2019). Critical features for face recognition. *Cognition*, *182*, 73–83. https://doi.org/10.1016/j.cognition.2018.09.002

Adams, R. B., & Weisbuch, M. (2026). A Case for Moving Beyond Verbal Labeling in Facial Expression Perception Research. *Affective Science*. https://doi.org/10.1007/s42761-025-00341-w

Adolphs, R., Nummenmaa, L., Todorov, A., & Haxby, J. V. (2016). Data-driven approaches in the investigation of social perception. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *371*(1693), Article 1693. https://doi.org/10.1098/rstb.2015.0367

Baltrusaitis, T., Zadeh, A., Lim, Y. C., & Morency, L.-P. (2018). OpenFace 2.0: Facial Behavior Analysis Toolkit. *2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018)*, 59–66. https://doi.org/10.1109/FG.2018.00019

Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. (2019). Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest: A Journal of the American Psychological Society*, *20*(1), Article 1. https://doi.org/10.1177/1529100619832930

Baudouin, J.-Y., Gallian, F., Pinoit, J.-M., & Damon, F. (2025). Arousal, valence, and discrete categories in facial emotion. *Scientific Reports*, *15*, 40268. https://doi.org/10.1038/s41598-025-24032-5

Betz, N., Hoemann, K., & Barrett, L. F. (2019). Words are a context for mental inference. *Emotion*, *19*(8), Article 8. https://doi.org/10.1037/emo0000510

Bockes, A., Hebart, M. N., & Lingnau, A. (2025). Revealing Key Dimensions Underlying the Recognition of Dynamic Human Actions. *Communications Psychology*, *3*(1), 149. https://doi.org/10.1038/s44271-025-00338-y

Bruce, V., Ellis, H. D., Gibling, F., & Young, A. (1987). Parallel processing of the sex and familiarity of faces. *Canadian Journal of Psychology / Revue Canadienne de Psychologie*, *41*(4), 510–520. https://doi.org/10.1037/h0084165

Bruce, V., & Young, A. (1986). Understanding face recognition. *British Journal of Psychology*, *77*(3), Article 3. https://doi.org/10.1111/j.2044-8295.1986.tb02199.x

Buckland, C. B., & Taubert, J. (2025). A database of naturalistic expressive faces for studying high arousal states. *The Journal of Pain*, *26*, 104728. https://doi.org/10.1016/j.jpain.2024.104728

Burton, A. M. (2013). Why has research in face recognition progressed so slowly? The importance of variability. *Quarterly Journal of Experimental Psychology*, *66*(8), 1467–1485. https://doi.org/10.1080/17470218.2013.800125

Chanes, L., Wormwood, J. B., Betz, N., & Barrett, L. F. (2018). Facial Expression Predictions as Drivers of Social Perception. *Journal of Personality and Social Psychology*, *114*(3), 380–396. https://doi.org/10.1037/pspa0000108

Cowen, A. S., & Keltner, D. (2020). What the Face Displays: Mapping 28 Emotions Conveyed by Naturalistic Expression. *The American Psychologist*, *75*(3), Article 3. https://doi.org/10.1037/amp0000488

Dawel, A., Krumhuber, E. G., & Palermo, R. (2025). Faking It Isn't Making It: Research Needs Spontaneous and Naturalistic Facial Expressions. *Affective Science*. https://doi.org/10.1007/s42761-025-00320-1

de Leeuw, J. R. (2015). jsPsych: A JavaScript library for creating behavioral experiments in a Web browser. *Behavior Research Methods*, *47*(1), Article 1. https://doi.org/10.3758/s13428-014-0458-y

Ekman, P., & Friesen, W. V. (1978). Facial action coding system. *Environmental Psychology & Nonverbal Behavior*.

Fujimura, T., Matsuda, Y.-T., Katahira, K., Okada, M., & Okanoya, K. (2012). Categorical and dimensional perceptions in decoding emotional facial expressions. *Cognition & Emotion*, *26*(4), 587–601. https://doi.org/10.1080/02699931.2011.595391

Grootswagers, T., Robinson, A. K., & Carlson, T. A. (2019). The representational dynamics of visual objects in rapid serial visual processing streams. *NeuroImage*, *188*, 668–679. https://doi.org/10.1016/j.neuroimage.2018.12.046

Grootswagers, T., Robinson, A. K., Shatek, S. M., & Carlson, T. A. (2024). Mapping the dynamics of visual feature coding: Insights into perception and integration. *PLOS Computational Biology*, *20*(1), Article 1. https://doi.org/10.1371/journal.pcbi.1011760

Hebart, M. N., Zheng, C. Y., Pereira, F., & Baker, C. I. (2020). Revealing the multidimensional mental representations of natural objects underlying human similarity judgments. *Nature Human Behaviour*, *4*(11), Article 11. https://doi.org/10.1038/s41562-020-00951-3

Hehman, E., Sutherland, C. A. M., Flake, J. K., & Slepian, M. L. (2017). The unique contributions of perceiver and target characteristics in person perception. *Journal of Personality and Social Psychology*, *113*(4), 513–529. https://doi.org/10.1037/pspa0000090

Henry, J. D., Grainger, S. A., & von Hippel, W. (2023). Determinants of Social Cognitive Aging: Predicting Resilience and Risk. *Annual Review of Psychology*, *74*(1), Article 1. https://doi.org/10.1146/annurev-psych-033020-121832

Holler, J. (2025). Facial clues to conversational intentions. *Trends in Cognitive Sciences*, *0*(0). https://doi.org/10.1016/j.tics.2025.03.006

Kriegeskorte, N., Mur, M., & Bandettini, P. (2008). Representational similarity analysis—Connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, *2*. https://www.frontiersin.org/articles/10.3389/neuro.06.004.2008

Kruskal, J. B., & Wish, M. (1978). *Multidimensional Scaling*. SAGE.

Leung, T. S., Maylott, S. E., Zeng, G., Nascimben, D. N., Jakobsen, K. V., & Simpson, E. A. (2023). Behavioral and physiological sensitivity to natural sick faces. *Brain, Behavior, and Immunity*, *110*, 195–211. https://doi.org/10.1016/j.bbi.2023.03.007

Lin, C., Keles, U., Thornton, M. A., & Adolphs, R. (2025). How trait impressions of faces shape subsequent mental state inferences. *Nature Human Behaviour*, *9*(1), 208–226. https://doi.org/10.1038/s41562-024-02059-4

Lindquist, K. A., & Gendron, M. (2013). What's in a Word? Language Constructs Emotion Perception. *Emotion Review*, *5*(1), 66–71. https://doi.org/10.1177/1754073912451351

Liu, M., Duan, Y., Ince, R. A. A., Chen, C., Garrod, O. G. B., Schyns, P. G., & Jack, R. E. (2022). Facial expressions elicit multiplexed perceptions of emotion categories and dimensions. *Current Biology*, *32*(1), 200-209.e6. https://doi.org/10.1016/j.cub.2021.10.035

Long, H., Peluso, N., Baker, C. I., Japee, S., & Taubert, J. (2023). A database of heterogeneous faces for studying naturalistic expressions. *Scientific Reports*, *13*(1), Article 1. https://doi.org/10.1038/s41598-023-32659-5

Matsumoto, D., & Wilson, M. (2026). What's in a Facial Stimulus? *Affective Science*. https://doi.org/10.1007/s42761-025-00342-9

Muttenthaler, L., Zheng, C. Y., McClure, P., Vandermeulen, R. A., Hebart, M. N., & Pereira, F. (2022). *VICE: Variational Interpretable Concept Embeddings* (arXiv:2205.00756; Issue arXiv:2205.00756). arXiv. https://doi.org/10.48550/arXiv.2205.00756

Nelson, N. L., & Russell, J. A. (2013). Universality Revisited. *Emotion Review*, *5*(1), Article 1. https://doi.org/10.1177/1754073912457227

Nili, H., Wingfield, C., Walther, A., Su, L., Marslen-Wilson, W., & Kriegeskorte, N. (2014). A Toolbox for Representational Similarity Analysis. *PLOS Computational Biology*, *10*(4), e1003553. https://doi.org/10.1371/journal.pcbi.1003553

Norton, R. W., & Pettegrew, L. S. (1979). Attentiveness as a style of communication: A structural analysis. *Communication Monographs*, *46*(1), Article 1. https://doi.org/10.1080/03637757909375987

Parkhi, O., Vedaldi, A., & Zisserman, A. (2015). Deep face recognition. *BMVC 2015 - Proceedings of the British Machine Vision Conference 2015*. https://ora.ox.ac.uk/objects/uuid:a5f2e93f-2768-45bb-8508-74747f85cad1

Peluso, N., Saurels, B. W., & Taubert, J. (2025). Toward ecological validity in expression discrimination: Forced-choice saccadic responses to posed and naturalistic faces. *Emotion*, *25*(7), 1872–1878. https://doi.org/10.1037/emo0001527

Porcheron, A., Mauger, E., & Russell, R. (2013). Aspects of Facial Contrast Decrease with Age and Are Cues for Age Perception. *PLOS ONE*, *8*(3), Article 3. https://doi.org/10.1371/journal.pone.0057985

Rossion, B. (2002). Is sex categorization from faces really parallel to face recognition? *Visual Cognition*, *9*(8), 1003–1020. https://doi.org/10.1080/13506280143000485

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, *39*(6), 1161–1178. https://doi.org/10.1037/h0077714

Saurels, B. W., Peluso, N., & Taubert, J. (2024). A behavioral advantage for the face pareidolia illusion in peripheral vision. *Scientific Reports*, *14*(1), Article 1. https://doi.org/10.1038/s41598-024-60892-z

Scarantino, A., Hareli, S., & Hess, U. (2022). Emotional expressions as appeals to recipients. *Emotion (Washington, D.C.)*, *22*(8), Article 8. https://doi.org/10.1037/emo0001023

Sutherland, C. A. M., & Young, A. W. (2022). Understanding trait impressions from faces. *British Journal of Psychology*, *113*(4), 1056–1078. https://doi.org/10.1111/bjop.12583

Taubert, J., & Alais, D. (2011). Identity aftereffects, but not composite effects, are contingent on contrast polarity. *Perception*, *40*(4), 422–436. https://doi.org/10.1068/p6874

Taubert, J., Alais, D., & Burr, D. (2016). Different coding strategies for the perception of stable and changeable facial attributes. *Scientific Reports*, *6*(1), 32239. https://doi.org/10.1038/srep32239

Taubert, J., Japee, S., Robinson, A., Long, H., Grootswagers, T., Zheng, C., Pereira, F., & Baker, C. (2024). The multidimensional representation of facial attributes. *Journal of Vision*, *24*(10), Article 10. https://doi.org/10.1167/jov.24.10.662

Taubert, J., Wally, S., & Dixson, B. J. (2023). Preliminary evidence of an increased susceptibility to face pareidolia in postpartum women. *Biology Letters*, *19*(9), 20230126. https://doi.org/10.1098/rsbl.2023.0126

Todorov, A., Dotsch, R., Wigboldus, D. H. J., & Said, C. P. (2011). Data-driven Methods for Modeling Social Perception. *Social and Personality Psychology Compass*, *5*(10), Article 10. https://doi.org/10.1111/j.1751-9004.2011.00389.x

Towler, A., Dunn, J. D., Castro Martínez, S., Moreton, R., Eklöf, F., Ruifrok, A., Kemp, R. I., & White, D. (2023). Diverse types of expertise in facial recognition. *Scientific Reports*, *13*(1), 11396. https://doi.org/10.1038/s41598-023-28632-x

Tsantani, M., Kriegeskorte, N., Storrs, K., Williams, A. L., McGettigan, C., & Garrido, L. (2021). FFA and OFA Encode Distinct Types of Face Identity Information. *Journal of Neuroscience*, *41*(9), 1952–1969. https://doi.org/10.1523/JNEUROSCI.1449-20.2020

Valentine, T., Lewis, M. B., & Hills, P. J. (2016). Face-Space: A Unifying Concept in Face Recognition Research. *Quarterly Journal of Experimental Psychology*, *69*(10), Article 10. https://doi.org/10.1080/17470218.2014.990392

Wardle, S. G., Paranjape, S., Taubert, J., & Baker, C. I. (2022). Illusory faces are more likely to be perceived as male than female. *Proceedings of the National Academy of Sciences*, *119*(5), Article 5. https://doi.org/10.1073/pnas.2117413119

Wei, Z., Robinson, A. K., Pegna, A. J., & Taubert, J. (2026). Parallel processing of distinct facial signals for the rapid evaluation of social agents. *iScience*, 114978. https://doi.org/10.1016/j.isci.2026.114978

White, D., & Burton, A. M. (2022). Individual differences and the multidimensional nature of face perception. *Nature Reviews Psychology*, *1*(5), Article 5. https://doi.org/10.1038/s44159-022-00041-3

Winkler, A. M., Ridgway, G. R., Webster, M. A., Smith, S. M., & Nichols, T. E. (2014). Permutation inference for the general linear model. *NeuroImage*, *92*, 381–397. https://doi.org/10.1016/j.neuroimage.2014.01.060

Yue, X., Biederman, I., Mangini, M. C., Malsburg, C. von der, & Amir, O. (2012). Predicting the psychophysical similarity of faces and non-face complex shapes by image-based measures. *Vision Research*, *55*, 41–46. https://doi.org/10.1016/j.visres.2011.12.012

Zhou, X., Vyas, S., Ning, J., & Moulson, M. C. (2022). Naturalistic Face Learning in Infants and Adults. *Psychological Science*, *33*(1), 135–151. https://doi.org/10.1177/09567976211030630